

HAVRE ST PIERRE, QC, CanadaWMO: **715840**Lat: **50.282N**Lon: **63.611W**Elev: **38**StdP: **100.87**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.3	(c) -23.8	(d) -32.6	(e) 0.2	(f) -25.7	(g) -30.0	(h) 0.2	(i) -23.0	(j) 15.8	(k) -6.5	(l) 13.7	(m) -6.4	(n) 3.6	(o) 280	(p) 0.752

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	22.8	17.1	21.4	16.5	20.1	15.7	18.6	21.2	17.6	19.9	16.7	19.0	4.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.6	19.6	16.6	11.9	18.8	15.7	11.2	17.9	52.8	21.2	49.9	20.1	47.2	19.0	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.5	9.3	-31.0	26.3	2.5	1.2	-32.8	27.1	-34.3	27.8	-35.7	28.5	-37.5	29.4
			-31.2	20.6	2.5	1.0	-33.0	21.4	-34.4	22.0	-35.8	22.6	-37.6	23.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-12.7	-11.8	-6.7	0.0	6.2	11.8	15.4	14.9	10.4	4.9	-1.4	-8.4		
	DBStd	10.69	6.40	6.35	5.76	3.48	2.99	2.91	2.19	2.32	3.07	3.13	4.63	6.05		
	HDD10.0	3372	702	611	518	301	123	14	0	1	31	160	341	571		
	HDD18.3	5980	961	844	777	551	375	197	95	107	238	416	591	829		
	CDD10.0	438	0	0	0	0	6	67	167	153	43	2	0	0		
	CDD18.3	5	0	0	0	0	0	0	3	2	1	0	0	0		
	CDH23.3	26	0	0	0	0	0	2	15	8	1	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	4.3	4.8	4.7	4.8	4.6	4.1	4.1	3.7	3.6	3.9	4.2	4.3	4.8		
(15) Precipitation	PrecAvg	1059	76	59	78	63	78	87	108	95	103	121	110	97		
	PrecMax	1294	128	92	148	135	132	144	165	169	166	218	200	166		
	PrecMin	925	23	22	42	22	29	34	44	50	49	43	56	60		
	PrecStd	106	32	18	30	29	32	33	34	31	31	40	34	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.2	5.9	11.4	19.6	23.2	25.1	24.4	21.4	15.2	11.1	5.8		
		MCWB	2.8	2.9	2.8	5.9	13.3	16.5	18.9	18.9	16.4	12.1	10.3	5.2		
	2%	DB	1.2	2.2	3.8	8.4	16.1	20.9	23.1	22.4	18.8	13.0	8.2	3.7		
		MCWB	0.6	0.8	1.2	4.6	10.5	14.9	17.4	17.4	15.0	10.2	7.4	2.9		
	5%	DB	-0.3	0.2	2.3	6.4	13.8	19.2	21.7	21.1	17.1	11.2	6.4	2.1		
		MCWB	-0.8	-0.7	0.3	3.3	9.2	13.8	16.6	16.8	13.8	9.3	5.3	1.2		
	10%	DB	-2.4	-2.0	1.0	4.9	12.0	17.5	20.3	19.8	15.6	9.9	5.0	0.6		
		MCWB	-3.1	-3.1	-0.5	2.3	8.0	12.9	16.1	16.0	12.7	8.2	4.1	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	3.5	3.3	7.1	13.8	18.3	20.2	19.9	17.8	13.2	10.6	5.2		
		MCDB	3.6	4.0	5.0	9.8	17.6	21.3	23.6	22.8	19.9	14.6	10.8	5.4		
	2%	WB	0.7	0.9	1.8	5.2	11.3	16.2	18.7	18.5	15.9	11.1	7.6	3.0		
		MCDB	1.3	1.7	3.0	7.7	14.7	19.5	21.5	21.1	17.7	12.1	8.0	3.5		
	5%	WB	-0.8	-0.7	0.7	3.9	9.8	14.9	17.7	17.6	14.5	9.8	5.6	1.4		
		MCDB	-0.4	0.3	1.8	5.9	12.8	18.0	20.2	19.7	16.1	10.8	6.2	1.9		
	10%	WB	-3.2	-3.0	-0.3	2.7	8.6	13.6	16.8	16.8	13.4	8.5	4.2	0.0		
		MCDB	-2.5	-2.1	0.9	4.4	11.4	16.5	19.3	18.7	15.0	9.6	4.8	0.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.8	9.3	7.5	8.8	9.2	8.9	9.1	8.9	7.5	6.9	8.0		
		MCDBR	9.9	7.7	8.1	9.7	12.6	12.2	11.4	11.0	10.7	9.1	7.1	6.9		
		MCWBR	9.9	7.3	6.5	8.8	7.8	6.9	6.6	6.6	7.1	6.9	6.6	6.8		
		MCDBR	9.9	8.0	7.3	8.3	11.0	10.3	9.4	9.2	8.0	6.8	6.8	7.2		
	5% WB	MCWBR	10.0	7.9	6.4	6.3	7.3	6.6	6.1	6.1	6.8	6.8	6.8	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.271	0.295	0.334	0.363	0.381	0.415	0.379	0.353	0.331	0.291	0.271		
	taud		2.352	2.330	2.327	2.353	2.403	2.379	2.335	2.440	2.484	2.509	2.509	2.365		
	Ebn at Noon		786	878	915	910	894	877	838	855	844	799	757	718		
	Edh at Noon		63	84	101	112	112	115	119	102	88	72	56	55		
(44) All-Sky Solar Radiation	RadAvg		1.04	1.97	3.32	4.50	5.20	5.54	5.29	4.54	3.43	2.10	1.10	0.82		
	RadStd		0.13	0.18	0.33	0.40	0.31	0.53	0.49	0.20	0.22	0.14	0.10	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.59	N/A	N/A	+0.42	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	+1.24	N/A	+0.33	+0.36	N/A	N/A	N/A	N/A

Nomenclature: See separate page